Liouville-Type Results for Infinity Elliptic Equations

Gradient-Dependent and Hardy-Hénon Nonlinearities

Tran Minh Hau[1,2] and Huynh Trung Hieu[2,3]

hautm.cs@gmail.com, hthieu.ue@gmail.com

[1]University of Information Technology (UIT), VNU-HCM
[2]Vietnam Institute for Advanced Study in Mathematics (VIASM)
[3]Ho Chi Minh City University of Education (HCMUE)

Advisors: Prof. Hai-Long Dao (University of Kansas),
Prof. Dac-Tuan Ngo (CNRS), Dr. Tuan-Minh Nguyen (Monash University)

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Motivation: Why Liouville Theorems Matter?

- **Rigidity:** They assert that global solutions under certain growth or sign conditions must be constant or trivial.
- **Blow-up Analysis:** Essential for understanding singularity formation in bounded domains.
- **Classical Foundation:** The Lane-Emden equation $-\Delta u = u^p$ has a critical exponent $p_S = \frac{n+2}{n-2}$ depending on dimension n.
- **The Shift:** Moving from the Laplacian (Δ) to the highly degenerate **Infinity Laplacian** (Δ_∞).
- **Research Gap:** Lack of exhaustive nonexistence results when Δ_∞ is coupled with gradient terms and spatial weights.

Problem Setup

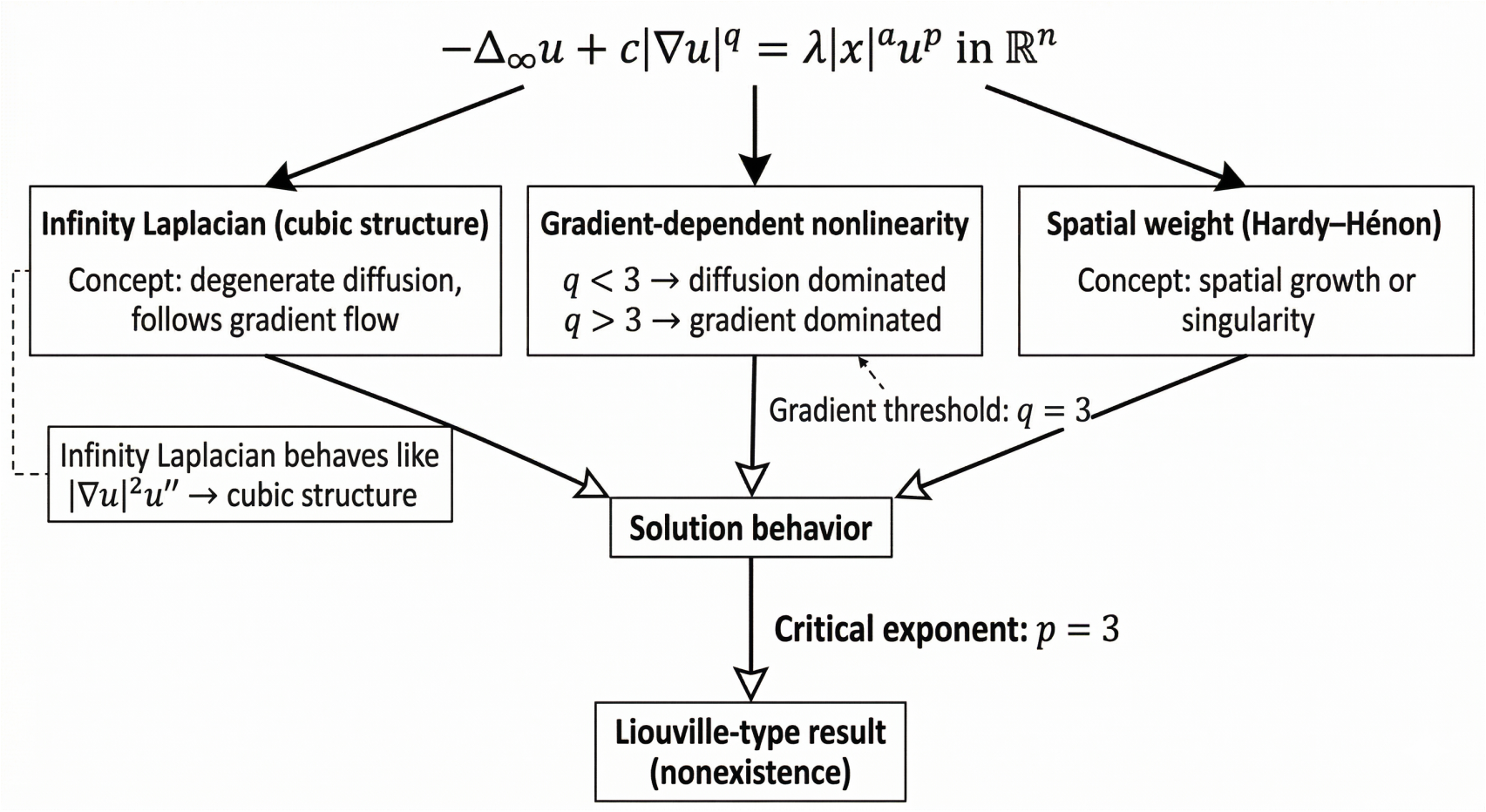
We investigate entire solutions $u \in C(\mathbb{R}^n)$ to the equation:

$$-\Delta_\infty u + c|\nabla u|^q = \lambda |x|^a u^p \quad \text{in } \mathbb{R}^n$$

Intuitive Breakdown:

- $\Delta_\infty u$: The "Infinity Diffusion" – second derivative along the gradient flow.
- $c|\nabla u|^q$: **Hamiltonian term** – acts as a gradient penalty ($c > 0$) or source ($c < 0$).
- $\lambda |x|^a u^p$: **Hardy-Hénon weight** – captures spatial heterogeneity ($a < 0$ singular, $a > 0$ growing).

Conceptual Structure of the Equation



Interaction between diffusion (infinity Laplacian), gradient nonlinearity, and Hardy–Hénon weight leading to Liouville-type nonexistence.

Key Challenges

- **Severe Degeneracy:** Δ_∞ vanishes wherever $\nabla u = 0$; the operator is non-uniformly elliptic.
- **Non-Divergence Form:** Traditional Sobolev (weak) solutions are inapplicable.
- **Viscosity Solutions:** We must work in the sense of Crandall, Ishii, and Lions (using test functions/superjets).
- **Competing Nonlinearities:** Tension between the cubic homogeneity of Δ_∞ and the first-order gradient absorption.

Main Results: The Critical Exponent

Theorem 3.1: Pure Hardy-Hénon Equation (c

Let $u \geq 0$ be a solution to $-\Delta_\infty u = \lambda |x|^a u^p$ with $a > -4$. If:

$$1 < p \leq \mathbf{3}$$

then $u \equiv 0$ is the only bounded solution.

- **Dimension Independence:** Unlike the standard Laplacian, the threshold $p^* = 3$ is independent of n.
- **Geometric Rigidity:** This reflects the localized, 1D nature of the infinity Laplacian scaling.

Main Results: Gradient Domination

Theorem 3.3: Bifurcation of Behavior

For $-\Delta_\infty u + c|\nabla u|^q \geq \lambda |x|^a u^p$ with $c > 0$, the system is split by $q_c = 3$:

1. **Diffusion Dominated ($q < 3$):** Collapse occurs if $p \leq 3$.
2. **Gradient Dominated ($q > 3$):** If $p < q$, solution collapses to $u \equiv 0$ under mild decay.

Theorem 3.4: Gradient Source

For $\Delta_\infty u + |c||\nabla u|^q + \lambda |x|^a u^p \leq 0$, **no positive subsolutions exist** for any $q > 0$ and $p \leq 3$.

The Intuition Slide: Why 3?

Conceptual Explanation:

- **Homogeneity:** The unnormalized Δ_∞ is cubic in terms of scaling: $\Delta_\infty(\tau u) = \tau^3 \Delta_\infty u$.
- **Classical vs. Infinity:**
  - **Laplacian:** Averages over a ball (n-dimensional volume effect).
  - **Infinity Laplacian:** Strictly follows gradient flow lines (1D geometric effect).
- **Energy Balance:** The "push" from the source $|x|^a u^p$ is countered by the "stiffness" of the operator. At $p \leq 3$, the source is too weak to sustain a global non-trivial shape without breaking the rigidity of the ∞-Laplacian.

Proof Sketch: Scaling and Radial Reduction

Step 1: Scaling Analysis

- Dilation $u_\tau(x) = \tau^\theta u(\tau x)$ leads to the balance: $\theta(p-3) = a+4$.
- If $p < 3$, decaying solutions are algebraically impossible.

Step 2: Radial Reduction

- For $v(r)$, the PDE collapses to the ODE: $-(v')^2 v'' \geq \lambda r^a v^p$.
- For power-law decay $v \sim r^{-\beta}$, the LHS is **always negative**, while the RHS is **always positive** $\implies$ Contradiction!

Connecting the Dots: Adversarial Robustness

- **The Vulnerability:** Neural networks $f(x)$ are sensitive to small perturbations ϵ.
- **The Solution:** Bound the global Lipschitz constant L.
- **The Math:** This is equivalent to bounding $\|\nabla f(x)\|_\infty \leq L$.
- **AMLE Theory:** Minimizing the L^∞ norm of the gradient yields exactly $\Delta_\infty f = 0$.
- **Result:** A perfectly robust network approximates an **infinity-harmonic function**.

Model Collapse: A PDE Perspective

PDE Equation $\longleftrightarrow$ ML Loss Function

- **Fidelity Term** ($|x|^a u^p$): Drives the model to fit training data.
- **Regularizer Term** ($c|\nabla u|^q$): Adversarial smoothing.
- **Insight from Theorem 3.3:**
  - If $q > 3$ (too aggressive regularization), the system collapses to $u \equiv 0$.
  - **Translation:** Over-regularizing a neural network forces it to ignore data entirely, leading to a "flat" non-predictive model.

Novel Contributions

- **Critical Exponents:** Identified $p^* = 3$ and $q_c = 3$ for the unnormalized operator.
- **Weighted Framework:** Developed new comparison principles for singular $|x|^a$ weights.
- **Generalizations:** Extended results to Matukuma-type and Exponential nonlinearities.
- **Interdisciplinary Link:** Provided a quantitative bound for "model collapse" in robust deep learning.

Conclusion and Future Directions

Summary:

- We established sharp nonexistence thresholds for infinity elliptic equations.
- The unnormalized operator's cubic homogeneity dictates a dimension-independent critical exponent of 3.

Open Problems:

- **Sign-Changing Solutions:** What happens when u is not strictly positive?
- **Anisotropic Weights:** How do critical exponents shift with non-radial potentials $w(x)$?

Thank You!